SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1510

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Code of Conduct

I K. Saipriya (192421296) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student K. Saipriya

CSA1510

CLOUD COMPUTING

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Assessment Tool No: 1

Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Sol:-

DEFINITION: cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet on a pay-as-you use basis. It allows users to access resources anytime without owning or maintaining physical infrastructure.

CHARACTERISTICS:

1. On-Demand Self-Service

* Users can create or remove computing resources whenever needed without contacting the service provider.

Example: Launching a virtual machine on AWS in a few minutes.

2. Broad Network Access

* Cloud services are accessible through the Interne[t] using devices like laptop, Smartphones, and tab[lets] from anywhere.

Example: Accessing Google Drive from any device

3. Resource pooling

* Cloud providers use shared servers and storage to serve multiple users securely through virtualization.

Example: Microsoft Azure shares Data center resources among many customers.

4. Rapid Elasticity

* Resources can be increased or decreased automatically based on demand.

Example: An e-commerce website increases server capacity during festival sales and reduce it afterward.

Difference from Traditional on-premise computing:

unlike traditional on-premise computing, where organizations must purchase, install, and maintain their own hardware, cloud computing provides scalable resources over the internet with lower upfront cost and easier management.

Conclusion:

cloud computing offers flexibility, scalability and cost efficiency, making it a better choice than traditional on-premise computing for many modern applications.

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2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platform

Sol:

Evolution of cloud computing

1. Hardware Evolution

* Early computing used large mainframe computers shared by many users.

* Later, personal computers (PCs) became common, giving users their own computing systems.

5. Mod

2. Network Evolution

* computers were connected through local area Network (LANs) and later the internet, enabling communication and resource sharing across different locations.

3. Virtualization

* Virtualization technology allowed multiple Virtual Machines (VMs) to run on a single physical server.

* This improved hardware utilization, reduced costs, and made resource management easier.

4. Internet Software Evolution

* Software shifted from being installed on individual computers to being delivered through the internet as web applications and services

Example: Gmail, google Docs, Microsoft 365.

5. Modern cloud platforms.

* combining virtualization, high speed internet, and web based software led to modern cloud platforms.

* Cloud provider such as AWS, Microsoft, now offer scalable, on-demand services like Iaas, Paas, Saas with a pay-as-you use model.

Flow of Evolution:

Mainframes → Personal computers → Internet & Networking → Virtualization → web applications → Modern cloud platforms (Aws, Azure, google cloud)

Conclusion:

The evolution from hardware-based computing to internet based software, supported by virtualization and networking, enabled today's cloud platforms to provide scalable, flexible, and cost-effective computing services

3) Explain the role of Server Virtualization in cloud computing. Differentiate between Virtualization as enabling technology and cloud computing as a Service model.

Sol:

Role of Server Virtualization in cloud computing:

Server Virtualization is the process of creating Multiple virtual Machines (VMs) on a single physical Server using a hypervisor. Each VM works like an independent computer with its own operating system and applications.

Role in cloud computing:

* Improves Resource utilization by allowing multiple VMs to share one Physical Server.

* Reduces hardware and maintenance costs.

* Provides Scalability as VMs can be created, removed or resized quickly.

* Supports isolation and Security where problems in one VM do no affect others

Example: A cloud provider like AWS runs many customer VMs on the same physical server using virtualization.

Difference between virtualization and cloud computing:

Virtualization (Enabling Technology)	Cloud computing (Service Model)
Creates multiple virtual machines on one physical server.	Delivers computing services over the internet.
Focuses on efficient use of hardware resources.	Focuses on providing services such as IaaS, PaaS, SaaS.
Works inside a server or data center.	Provides services to users on demand through the Internet.
Example: VMware, Hyper-V	Example: AWS, Microsoft Azure, google cloud

Conclusion: Virtualization is the technology that enables efficient use of hardware, while cloud computing is the service model that uses virtualization to deliver scalable & on-demand computing services over the internet.

4 Discuss the role of Data centers in delivering cloud services.

Sol:

Definition:

A data center is a facility that contains servers, storage systems, networking devices, and cooling and power systems. It is the physical infrastructure where cloud services are hosted and managed.

Role of Data centers in cloud services:

1. Resource Hosting

* Stores and runs cloud resources such as virtual machines, databases, and applications

Example: AWS data centers host EC2 instances

2. Data Storage and Backup

* Stores user data securely and creates backups to prevent data loss

Example: Google cloud storage.

3. High Availability

* uses redundant power, cooling, and network connections to ensure service are available 24/7 with minimal downtime.

4. Scalability

* Allows cloud providers to add or remove computing resources based on user demand.

5. Security

* protects data through physical security (CCTV, biometric access) and cyber security measures (firewalls, encryption).

Conclusion:

Data centers are the backbone of cloud computing. They provide the infrastructure needed to deliver reliable, scalable, secure, and high-performance cloud services to users around the world.

Q5. Differentiate Iaas, Paas, Saas and Xaas with one suitable real-world example for each model

Service Model	Definition	Real-world example
IaaS (infrastructure as a service)	Provides virtual servers, storage, and networking over the internet. User manage the OS & applications.	Amazon EC2 (AWS
PaaS (Platform as a Service)	Provides a platform for developing, testing, and deploying applications without managing hardware.	Google App Engine.
SaaS (Software as a Service)	Provides ready-to-use Software that users access through a web browser.	Google Docs
XaaS (Anything as a Service)	A broad model where any IT Service is delivered over the Internet on demand	Dropbox (Storage as a Service).

Example: ...

Key Difference:

* Iaas: Infrastructure is provided
* Paas: Platform for application development is provided
* Saas: Complete software application is provided.
* Xaas: Any IT Service can be delivered as a cloud service.

Conclusion:

Cloud Service models provide different levels of services based on user needs.

Iaas offers infrastructure, Paas offers a development platform, Saas offers ready-to-use software. and Xaas extends cloud services to almost any IT resource.